American University of Sharjah

College of Engineering

Department of Computer Engineering and Science

CMP320 – Database Systems

Final Project

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Due Date

April 28, 2025

Project Title:

The Airline Reservation and Management database system

Project Description:

The Airline Reservation and Management database system where airports, flights, and passengers operate to handle bookings, payments, luggage handling, and in-flight services. It enables flight search, seat booking, ticketing, processing payments, airport cooperation, luggage tracking, and in-flight services. Passengers may look for available flights, reserve seats, and pay online, while administrators can plan and monitor flights, allocate crews, and handle airport departures. The schema maintains data integrity and consistency using standardized relational tables, strong primary/foreign key constraints, and allows for composite and multi-valued characteristics. Its integrated approach supports complicated operational and reporting inquiries.

- **Airports** are identifiable by a unique airport_id and specified by their name, iata code, country, city, and time zone.
- **Flights** are arranged by airlines and assigned by a unique flight_id. Each flight data include airline, departure and arrival times, a derived duration ((arrival time – departure time)×24), aircraft model and status.
- **Passengers** are assigned a unique passenger_id and provide their passport number, first name, middle initial, last name, date of birth, gender, nationality, address street, city, zip, and email. A view passenger_with_age calculates each passenger's current age using date of birth.
- **Tickets** are provided to passengers for flights. Each ticket has a unique ticket_id and records class, ticket date, ticket time, price, seat number, gate, boarding time, and guarantees that every issued ticket belongs to an existing passenger and corresponds to a scheduled flight. An optional payment_id links to the corresponding payment.
- **Payments** are tracked one-to-one with tickets: every payment applies to exactly one ticket, and no ticket can be paid more than once. Payments data also include amount, method, and date of payment.
    - Each passenger may have zero or more phone number and zero or more preference entries.

- **Staff** members (pilots, attendants, crew) are specified by staff_id and described by their first name, last name, email, phone, and role. Each is allocated to exactly one flight using flight_id.
- **FlightDepartures** captures the M:N relationship between Airport and Flight. An airport can handle many flights, and a flight may depart from several airports.
- **Luggage** belong to both a ticket and its passenger. Each luggage_id entry contains a tag number, weight, type (Checked/Carry-on), color, flags if fragile and if overweight, and status (e.g., Loaded, In Transit, Delivered).
- **Services** links passenger receiving it, the flight on which it occurs, and the crew member who provides it. It keeps track of which staff member performed what sort of service type (e.g., Meal Delivery, Safety Demonstration) to which passenger on which flight and at what time.

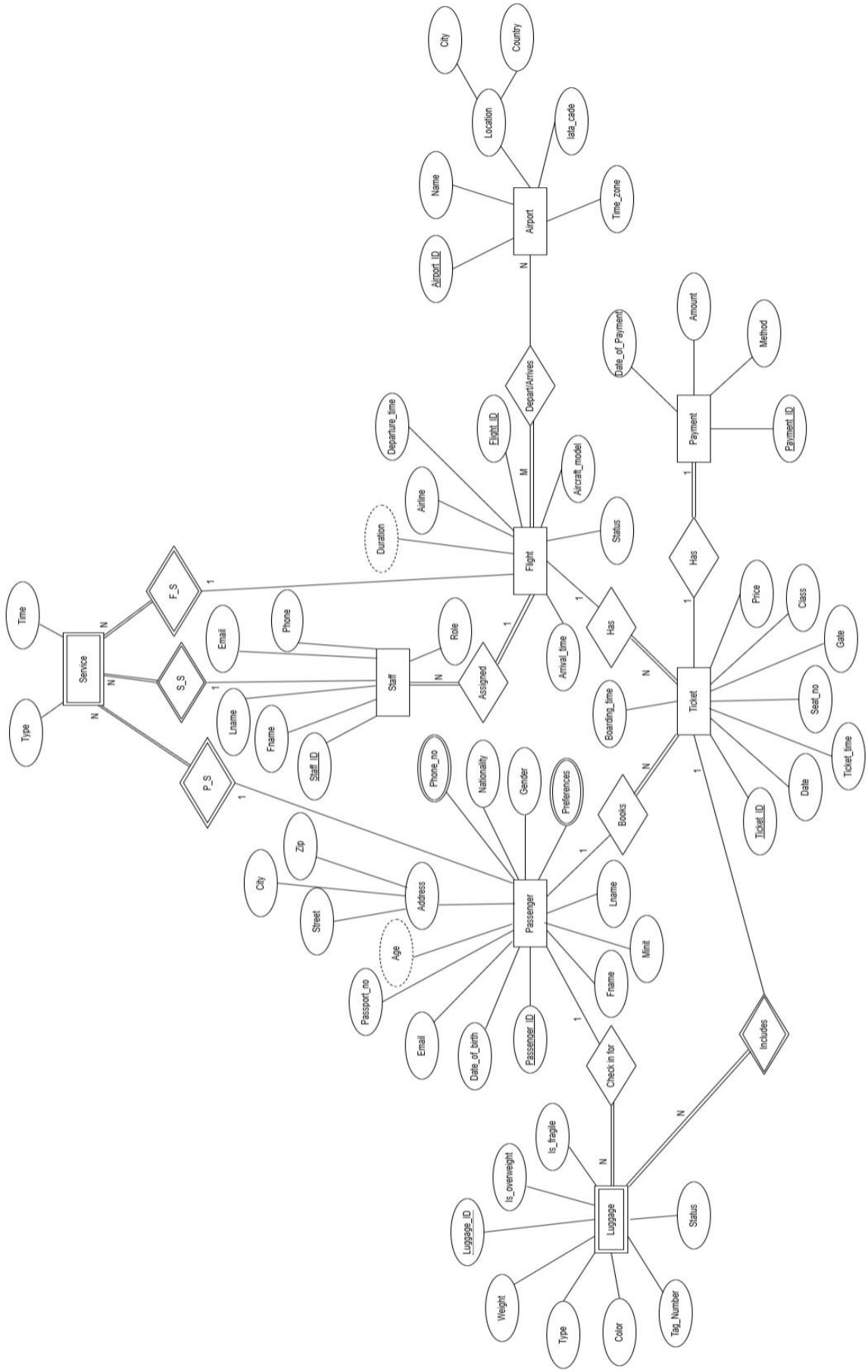
Keys and Constraints

1. **Airport**
    - **Primary Key (PK):** airport_id
    - **Composite Attribute:** location (country, city)
    - **Attributes:** airport_name, iata_code, time_zone
2. **Flight**
    - **Primary Key (PK):** flight_id
    - **Derived Attribute:** duration (computed as (arrival_time – departure_time) × 24)
    - **Attributes:** airline, departure_time, arrival_time, aircraft_model, status
3. **Passenger**
    - **Primary Key (PK):** passenger_id
    - **Unique Constraint:** passport_no
    - **Composite Attribute:** address (address_street, address_city, address_zip)
    - **Multi-valued Attributes:** phone_number, preference
    - **Derived Attribute:** age (from date_of_birth using view)
    - **Attributes:** first_name, middle_name_initial, last_name, date_of_birth, gender, nationality, email
4. **Ticket**
    - **Primary Key (PK):** ticket_id
    - **Unique Constraint:** payment_id
    - **Foreign Keys (FK):**
        - passenger_id → passenger1(passenger_id)
        - flight_id → flight1(flight_id)
        - payment_id → payment1(payment_id)
    - **Attributes:** class, ticket_date, ticket_time, price, seat_number, gate, boarding_time

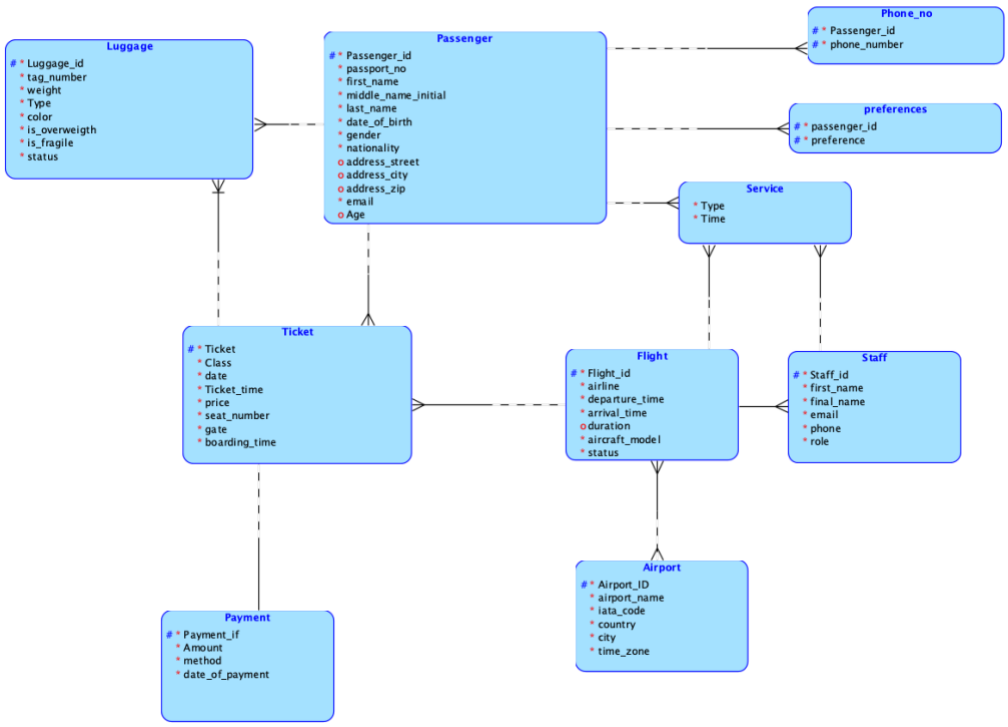
5. **Payment**
    - **Primary Key (PK):** payment_id
    - **Unique Constraint:** ticket_id
    - **Foreign Key (FK):**
        - ticket_id → ticket1(ticket_id)
    - **Attributes:** amount, method, date_of_payment
6. **Phone Number** (multi-valued)
    - **Composite PK:** (passenger_id, phone_number)
    - **Foreign Key (FK):**
        - passenger_id → passenger1(passenger_id)
7. **Preferences** (multi-valued)
    - **Composite PK:** (passenger_id, preference)
    - **Foreign Key (FK):**
        - passenger_id → passenger1(passenger_id)
8. **Staff**
    - **Primary Key (PK):** staff_id
    - **Foreign Key (FK):**
        - flight_id → flight1(flight_id)
    - **Attributes:** first_name, last_name, email, phone, role
9. **FlightDepartures** (M:N)
    - **Composite PK:** (airport_id, flight_id)
    - **Foreign Keys (FK):**
        - airport_id → airport1(airport_id)
        - flight_id → flight1(flight_id)
10. **Luggage** (weak entity)
    - **Composite PK:** (luggage_id, ticket_id)
    - **Foreign Keys (FK):**
        - ticket_id → ticket1(ticket_id)
        - passenger_id → passenger1(passenger_id)
    - **Attributes:** tag_number, weight, type, color, is_fragile, is_overweight, status

11. **Service** (ternary relationship)
    - **Composite PK:** (passenger_id, flight_id, staff_id)
    - **Foreign Keys (FK):**
        - passenger_id → passenger1(passenger_id)
        - flight_id → flight1(flight_id)
        - staff_id → staff1(staff_id)
    - **Attributes:** service_type, service_time

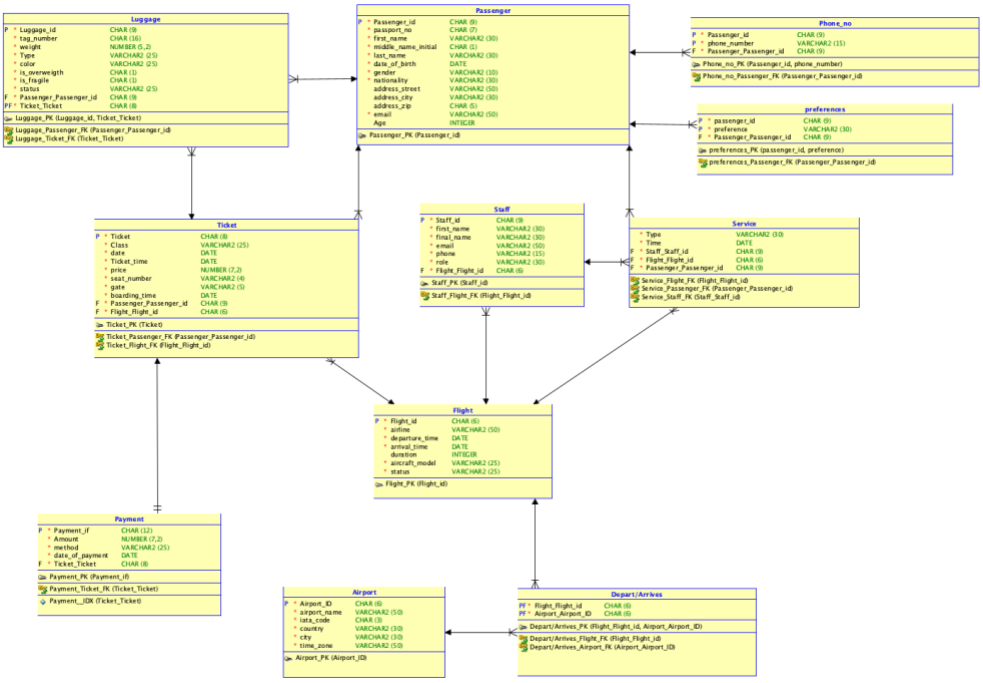
ER Diagram:


Service
Time
Type
F_S
S_S
P_S
Staff
Email
Phone
Lname
Fname
Staff_ID
Role
Assigned
Flight
Departure_time
Airline
Duration
Flight_ID
Aircraft_model
Status
Arrival_time
Depart/Arrives
Airport
City
Country
Location
Iata_cade
Name
Airport_ID
Time_zone
Has
Payment
Amount
Method
Date_of_Payment
Payment_ID
Ticket
Price
Class
Gate
Seat_no
Ticket_time
Date
Ticket_ID
Boarding_time
Books
Passenger
Phone_no
Nationality
Gender
Preferences
Address
Zip
City
Street
Age
Passport_no
Email
Date_of_birth
Passenger_ID
Lname
Minit
Fname
Check in for
Includes
Luggage
Is_fragile
Is_overweight
Luggage_ID
Weight
Type
Color
Tag_Number
Status

Logical Diagram:



Relational Diagram:



ER to Relational Mapping:

Queries:

SELECT *FROM PASSENGER;

	PASSENGER_ID	PASSP...	FIR...	MIDDLE_NAME_INITIAL	LAST_NAME	DATE_OF_BIRTH	GENDER	NATIONALITY	ADDRESS_STREET	ADDRE...	ADDRE...	EMAIL
1	PA0000002	B234567	Bob	B	Brown	02/02/85	Male	United Kingdom	456 Oak Rd	London	00002	bob@example.co.uk
2	PA0000003	C345678	Carol	C	Clark	03/03/92	Female	United Arab Emirates	789 Maple Ave	Dubai	00005	carol@example.com
3	PA0001001	G123456	Grace	G	Green	06/06/91	Female	British	12 King Rd	London	W1A1AB	grace.green@example.co.uk
4	PA0001002	H234567	Harry	H	Hill	07/07/89	Male	British	34 Queen St	London	W1A1AC	harry.hill@example.co.uk
5	PA0000001	A123456	Alice	B	Smith	15/01/95	Female	American	123 Main St	New York	10001	alice.smith@example.com

SELECT *FROM FLIGHT;

	FLIGHT_ID	AIRLINE	DEPARTURE_TIME	ARRIVAL_TIME	AIRCRAFT_MODEL	STATUS	DURATION
1	FL0000007	Cathay Pacific	13/05/25	13/05/25	Boeing 777	Scheduled	8.5
2	FL0000006	Singapore Airlines	12/05/25	12/05/25	Airbus A350	Scheduled	7.5
3	FL0000005	Qantas	11/05/25	11/05/25	Boeing 787	Scheduled	8
4	FL0000004	NoAir	04/05/25	04/05/25	ModelY	Scheduled	2
5	FL0000001	Emirates	01/05/25	01/05/25	Boeing 777	Scheduled	4
6	FL0000002	British Airways	02/05/25	02/05/25	Airbus A380	Scheduled	10
7	FL0000003	Qantas	10/05/25	10/05/25	Boeing 787	Delayed	8

SELECT *FROM TICKET;

	TICKET_ID	CLASS	TICKET_DATE	TICKET_TIME	PRICE	SEAT_NUMBER	GATE	BOARDING_TIME	PASSENGER_ID	FLIGHT_ID	PAYMENT_ID
1	TK000002	Business	21/04/25	21/04/25	200	1B	B2	02/05/25	PA0000001	FL0000002	PMT000000002
2	TK000003	First	22/04/25	22/04/25	50	1C	C3	03/05/25	PA0000001	FL0000003	PMT000000003
3	TK000004	Economy	23/04/25	23/04/25	150	22D	D4	04/05/25	PA0000002	FL0000001	PMT000000004
4	TK000005	Economy	24/04/25	24/04/25	120	23E	E5	05/05/25	PA0000003	FL0000001	PMT000000005
5	TK000001	TK000001	20/04/25	20/04/25	1200.5	12A	A1	01/05/25	PA0000001	FL0000001	PMT000000001

SELECT *FROM PAYMENT;

	PAYMENT_ID	AMOUNT	METHOD	DATE_OF_PAYMENT	TICKET_ID
1	PMT000000003	50	Card	22/04/25	TK000003
2	PMT000000004	150	Cash	23/04/25	TK000004
3	PMT000000002	200	Cash	21/04/25	TK000002
4	PMT000000005	120	Card	24/04/25	TK000005
5	PMT000000001	1200.5	Debit Card	20/04/25	TK000001

SELECT *FROM AIRPORT;

	AIRPORT_ID	AIRPORT_NAME	IATA_CODE	COUNTRY	CITY	TIME_ZONE
1	AP0003	Changi Airport	SIN	Singapore	Singapore	Asia/Singapore
2	AP0001	Dubai International	DXB	United Arab Emirates	Dubai	Asia/Dubai
3	AP0002	Heathrow Airport	LHR	United Kingdom	London	Europe/London

SELECT *FROM STAFF;

	STAFF_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE	ROLE	FLIGHT_ID
1	ST0000003	Jim	Beam	jim@example.com	+971500000012	Crew	FL0000002
2	ST0000001	John	Doe	john.doe@emirates.com	+971509876543	Pilot	FL0000001
3	ST0000002	Jane	Doe	jane.doe@emirates.com	+971509876544	Attendant	FL0000001

SELECT *FROM LUGGAGE;

	LUGGAGE_ID	TICKET_ID	PASSENGER_ID	TAG_NUMBER	WEIGHT	TYPE	COLOR	IS_FRAGILE	IS_OVERWEIGHT	STATUS
1	LG0000001	TK000001	PA0000001	TAG000000000001	23.5	Checked	Black	N	N	Loaded
2	LG0000002	TK000001	PA0000001	TAG000000000002	8	Carry-on	Red	N	N	In Transit

SELECT *FROM FLIGHT_DEPARTURES;

	AIRPORT_ID	FLIGHT_ID
1	AP0001	FL0000001
2	AP0001	FL0000002
3	AP0002	FL0000002
4	AP0002	FL0000003

SELECT *FROM PHONE_NUMBER;

	PASSENGER_ID	PHONE_NUMBER
1	PA0000001	+971500000001
2	PA0000001	+971501234567
3	PA0000003	+971500000003

SELECT *FROM PREFERENCES;

	PASSENGER_ID	PREFERENCE
1	PA0000001	Vegetarian
2	PA0000001	Window Seat

SELECT *FROM SERVICE;

	PASSENGER_ID	FLIGHT_ID	STAFF_ID	SERVICE_TYPE	SERVICE_TIME
1	PA0000001	FL0000001	ST0000002	Meal Delivery	01/05/25

```
--Equi-Join
SELECT p.first_name, t.ticket_id
  FROM passenger p
  JOIN ticket t
    ON p.passenger_id = t.passenger_id;
```

	FIRST_NAME	TICKET_ID
1	Bob	TK000004
2	Carol	TK000005
3	Alice	TK000001
4	Alice	TK000003
5	Alice	TK000002

```
--Natural Join
SELECT *
  FROM passenger
  NATURAL JOIN ticket;
```

	PASSEN...	PAS...	FIRS...	MI...	LA...	DATE_...	GENDER	NATI...	AD...	ADDR...	ADDR...	EM...	TI...	CLASS	TICKE...	TIC...	PRICE	S...	GATE	BOAR...	FLIG...	PAYMENT_ID
1	PA0000002	B234567	Bob	B	Brown	02/02/85	Male	United...	456 ...	London	00002	bob@...	TK...	Economy	23/04/25	23/...	150	22D	D4	04/05/25	FL000...	PMT000000004
2	PA0000003	C345678	Carol	C	Clark	03/03/92	Female	United...	789 ...	Dubai	00005	caro...	TK...	Economy	24/04/25	24/...	120	23E	E5	05/05/25	FL000...	PMT000000005
3	PA0000001	A123456	Alice	B	Smith	15/01/95	Female	American	123 ...	New York	10001	alic...	TK...	TK000001	20/04/25	20/...	1200.5	12A	A1	01/05/25	FL000...	PMT000000001
4	PA0000001	A123456	Alice	B	Smith	15/01/95	Female	American	123 ...	New York	10001	alic...	TK...	First	22/04/25	22/...	50	1C	C3	03/05/25	FL000...	PMT000000003
5	PA0000001	A123456	Alice	B	Smith	15/01/95	Female	American	123 ...	New York	10001	alic...	TK...	Business	21/04/25	21/...	200	1B	B2	02/05/25	FL000...	PMT000000002

```
--Outer join
SELECT f.flight_id, t.ticket_id
  FROM flight f
  LEFT OUTER JOIN ticket t
    ON f.flight_id = t.flight_id;
```

	FLIGHT_ID	TICKET_ID
1	FL0000002	TK000002
2	FL0000003	TK000003
3	FL0000001	TK000004
4	FL0000001	TK000005
5	FL0000001	TK000001
6	FL0000004	(null)
7	FL0000006	(null)
8	FL0000007	(null)
9	FL0000005	(null)

```
--Self-Join
SELECT p1.passenger_id AS p1_id, p1.first_name AS p1_name,
       p2.passenger_id AS p2_id, p2.first_name AS p2_name
  FROM passenger p1
  JOIN passenger p2
    ON p1.address_city = p2.address_city
   AND p1.passenger_id <> p2.passenger_id;
```

	P1_ID	P1_NAME	P2_ID	P2_NAME
1	PA0001002	Harry	PA0000002	Bob
2	PA0001001	Grace	PA0000002	Bob
3	PA0001002	Harry	PA0001001	Grace
4	PA0000002	Bob	PA0001001	Grace
5	PA0001001	Grace	PA0001002	Harry
6	PA0000002	Bob	PA0001002	Harry

```
--Subquery
SELECT flight_id, duration
  FROM flight
 WHERE duration > (
   SELECT AVG(duration)
     FROM flight
 );
```

	FLIGHT_ID	DURATION
1	FL0000007	8.5
2	FL0000006	7.5
3	FL0000005	8
4	FL0000002	10
5	FL0000003	8

```
--HAVING
SELECT passenger_id, COUNT(*) AS cnt
  FROM ticket
 GROUP BY passenger_id
HAVING COUNT(*) > (
    SELECT AVG(cnt)
      FROM (
        SELECT passenger_id, COUNT(*) AS cnt
          FROM ticket
         GROUP BY passenger_id
      )
);
```

	PASSENGER_ID	CNT
1	PA0000001	3

```
--IN
SELECT staff_id, role
  FROM staff
 WHERE flight_id IN (
   SELECT flight_id
     FROM flight_departures
    WHERE airport_id = 'AP0001'
 );
```

	PASSENGER_ID	CNT
1	PA0000001	3

```
--IN
SELECT DISTINCT p.passenger_id, p.first_name
  FROM passenger p
 WHERE p.nationality IN (
   SELECT country
    FROM airport
 );
```

	PASSENGER_ID	FIRST_NAME
1	PA0000003	Carol
2	PA0000002	Bob

```
--nested subqueries
SELECT *
  FROM payment
 WHERE amount > (
   SELECT AVG(amount)
    FROM payment
   WHERE ticket_id IN (
     SELECT ticket_id
      FROM ticket
     WHERE flight_id = (
      SELECT flight_id
       FROM flight
      WHERE airline = 'Emirates'
     )
   )
 );
```

	PAYMENT_ID	AMOUNT	METHOD	DATE_OF_PAYMENT	TICKET_ID
1	PMT000000001	1200.5	Debit Card	20/04/25	TK000001

```
--LEFT OUTER JOIN
SELECT
  p.passenger_id,
  p.first_name,
  pn.phone_number
FROM passenger p
LEFT OUTER JOIN phone_number pn
  ON p.passenger_id = pn.passenger_id;
```

	PASSENGER_ID	FIRST_NAME	PHONE_NUMBER
1	PA0000001	Alice	+971500000001
2	PA0000001	Alice	+971501234567
3	PA0000002	Bob	(null)
4	PA0000003	Carol	+971500000003
5	PA0001001	Grace	(null)
6	PA0001002	Harry	(null)

```
--RIGHT OUTER JOIN
SELECT
  pn.phone_number,
  p.passenger_id,
  p.first_name
FROM phone_number pn
RIGHT OUTER JOIN passenger p
  ON pn.passenger_id = p.passenger_id;
```

	PHONE_NUMBER	PASSENGER_ID	FIRST_NAME
1	+971500000001	PA0000001	Alice
2	+971501234567	PA0000001	Alice
3	(null)	PA0000002	Bob
4	+971500000003	PA0000003	Carol
5	(null)	PA0001001	Grace
6	(null)	PA0001002	Harry